

Derek C. Carter

Staff Cloud Infrastructure

Engineer

Platform Engineering Leader

Kubernetes & Cloud Architecture

Specialist

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Summary:

Staff-level Infrastructure Engineer with 15+ years designing and operating cloud platforms at scale. Expert in Kubernetes cluster architecture, infrastructure-as-code, orchestration, and observability for platform engineering that enables engineering teams to ship without ops dependencies. Deep experience in fintech environments with compliance controls (SOC2, SOX, PCI-DSS) and track record of setting technical direction through RFCs, architecture reviews, and cross-functional leadership. Scaled infrastructure supporting 100+ microservices across multiple AWS regions.

Experience

VidAngel (vidangel.com) – Principal DevOps Engineer – Jun 2025 - Feb 2026

Principal DevOps Engineer architecting scalable cloud infrastructure, Kubernetes cluster management, and zero-downtime deployment strategies for a streaming media company.

- Designed self-hosted HashiCorp Vault architecture with high-availability cluster, automated key management, and load balancing, achieving 99.95% service availability and delivering approximately 30% reduction in secret management costs.
- Architected zero-downtime backend deployment pipeline using GitLab CI/CD with rolling deployment strategies, implementing hash-based dependency packaging and automated rollback capabilities via Ansible for continuous deployment.
- Built automated golden image pipeline using Packer and Ansible, creating standardized AMI base operating system images with integrated monitoring, logging agents, and network security configurations across all environments.
- Developed and maintained custom Debian packaging (.deb) for internal tooling and service dependencies, enabling reproducible artifact delivery and streamlined OS-level installation across fleet environments.
- Deployed and managed self-hosted GitLab CI/CD runners on Kubernetes infrastructure, reducing external dependencies and improving build security with custom runner images containing required development tooling.
- Expanded Infrastructure as Code adoption through reusable Terraform modules for VPC networking, load balancers, and access management, establishing consistent deployment patterns and accelerating team onboarding.

Unchained (unchained.com) – Senior Software Engineer / DevOps Team Lead – May 2023 - Mar 2025

Staff-level DevOps Team Lead driving platform transformation and cloud infrastructure modernization for high-availability cryptocurrency platform with 50+ engineers.

- Established Request for Comments (RFC) process for engineering architecture decisions, setting technical direction and improving cross-team collaboration across microservices platform.
- Led GitLab Enterprise migration implementing GitOps deployment methodology with self-hosted GitLab server and enhanced security controls.
 - Architected enterprise-wide Kubernetes deployment platform with standardized CI/CD pipelines, supporting 30+ microservices across multiple AWS regions and enabling developer self-service capabilities.
 - Implemented private GitLab CI/CD runners on Kubernetes infrastructure, ensuring code security compliance and reducing third-party dependencies for continuous deployment.
- Designed and deployed multi-environment cloud architecture with environment-specific internal domains, load balancers, and network segmentation, reducing deployment errors and improving developer experience.
- Enhanced Terraform Infrastructure as Code with Terragrunt multi-environment architecture, enabling consistent infrastructure deployment across development, staging, and production with automated rollback capabilities.
- Implemented policy-as-code access controls using Terraform and Terragrunt to enforce Role Based Access Control (RBAC) across AWS environments, embedding identity and access management into platform abstractions and enforcing least-privilege by default.
- Wrote on-call runbooks and defined rotation schedules for the microservices platform; established repeatable post-incident review process and cut triage time for production issues.
- Managed team of 2 DevOps engineers and established rotating “DevOps Residency Program” to embed DevOps practices across 8 development teams, improving deployment frequency and system reliability.

UserTesting (usertesting.com) – Staff Software Engineer / Ops and Infrastructure Team Lead – Mar 2021 - Mar 2023

Staff-level infrastructure leader driving platform architecture for enterprise-scale user experience platform supporting 100+ microservices.

- Led 7-person infrastructure team with quarterly OKR planning, balancing technical debt reduction with new feature delivery across 100+ microservices.
- Established engineering best practices and led DevOps Guild and Cross-Squad Architecture Guild, standardizing zero-downtime deployment practices including rolling, canary, and blue/green strategies across 15 development teams.
- Ensured enterprise security compliance (SOC2, SOX, GDPR, ISO27001) through infrastructure automation and security tooling implementation, maintaining compliance audits.
- Architected modern container orchestration platform using Amazon Elastic Kubernetes Service (EKS), Terraform Infrastructure as Code, and GitOps deployment methodology.
- Migrated critical internal applications from Heroku to self-managed Amazon EKS Kubernetes infrastructure, reducing operational costs by 40% while improving security posture.
- Enhanced developer experience through comprehensive platform engineering initiatives:
 - Developed reusable GitHub Actions and CircleCI pipeline templates with GitOps integration, reducing CI/CD setup time for new microservices by 75%.
 - Created standardized Terraform IaC templates for rapid microservice deployment with auto-scaling and load balancing.
- Wrote runbooks and on-call escalation paths for the 100+ service platform, giving 15 development teams clear ownership during incidents and faster paths to recovery.
- Maintained and optimized AWS ECS legacy infrastructure supporting high-availability production workloads.

Nav (nav.com) – Principal Software Engineer DevOps – Aug 2017 - Feb 2021

Principal DevOps Engineer architecting cloud-based SaaS applications for fintech startup. Led complete platform transformation migrating traditional AWS infrastructure to Kubernetes cluster design with zero-downtime deployment strategies.

- Mitigated legacy architecture problems using Infrastructure as Code automation (Ansible, Terraform), security tools (HashiCorp Vault), and Linux systems operations expertise including network fundamentals and load balancing.
 - Created open-source [vault_helper](#) tool for automated rotation of encryption keys in HashiCorp Vault system, improving security automation.
- Architected and deployed production-grade Kubernetes platform with cluster design, CI/CD integration, and troubleshooting capabilities.
 - Built hybrid cloud architecture enabling seamless communication between legacy applications and modern Kubernetes workloads using advanced networking, load balancers, and service mesh technologies.
- Enhanced developer workflow processes and platform engineering tooling for improved scalability and reliability.
 - Established company-wide engineering standards including JSON logging, HashiCorp Vault secret management, and automated deployment workflows with observability.
 - Built comprehensive “canary application” for testing zero-downtime deployment strategies including rolling, canary, and blue/green deployments across all platform features.
 - Architected centralized CI/CD pipeline platform to standardize deployment workflows, enabling developer self-service capabilities and reducing deployment cognitive load.
 - Migrated entire infrastructure to HashiCorp Terraform Infrastructure as Code, establishing repeatable, maintainable, and version-controlled infrastructure with automated provisioning.
 - Wrote RunDocs and wired up observability for every service, giving the ops team actionable triage steps and cutting time-to-resolution on production incidents.

Goozbach Infrastructure Solutions, LLC (goozbach.com) – Linux Infrastructure Consultant and Instructor – Dec 2011 – Present

Owner and primary consultant of Linux consulting and training company providing Linux support and best practices consultation. Customers include Red Hat training partners, Linux Foundation, IBM, HP, Novell, Sangoma, Local, State, Federal Government, and the Department of Defense. Customer environments range from dozens to thousands of servers.

Services provided include system updates, configuration management, installation automation, Bash and Python scripting, training delivery, training course development, and systems administration tasks.

- Consulting customers on DevOps policies, procedures, and platform engineering tools, using continuous integration (Jenkins), configuration management (Puppet, Ansible, Salt), and installation automation (PXE, Kickstart)
- Infrastructure consulting for security startup in stealth mode
- Built and maintained **GooseOS**, a custom RHEL-derivative Linux distribution: authored RPM spec files, automated package builds with Koji, and managed custom Yum/DNF repositories for targeted deployment environments.
- Linux distribution customization for telephony-space client: modified upstream RHEL/CentOS and Debian/Ubuntu bases, packaged custom RPMs and `.deb` packages, and built automated installation workflows (PXE, Kickstart, Preseed, Autoinstall) for hardware provisioning at scale.
- Created Linux Courseware and virtual/hybrid classroom infrastructure
- Guided consulting clients through on-premises data center to AWS cloud migrations, including workload assessment, lift-and-shift execution, and hybrid architecture design bridging legacy Linux infrastructure with AWS services.

Technical Skills

Kubernetes Expertise: Cluster Design • CI/CD Integration • Service Mesh • Observability • Troubleshooting • Container Orchestration

Cloud Platforms & Infrastructure: AWS (EC2 • ECS • EKS • Lambda • RDS • S3) • GCP • Azure • Auto-scaling • Multi-Region Deployments • Cost Optimization

Infrastructure as Code: Terraform • Terragrunt • CloudFormation • Ansible • Packer

GitOps & CI/CD: ArgoCD • GitLab Pipelines • GitHub Actions • Jenkins • Zero-Downtime Deployments

Network Fundamentals: TCP/UDP • IPv4/IPv6 • Subnetting • OSI Model • Firewalls • Load Balancers • VPN • Service Discovery

Observability & Monitoring: CloudWatch • Prometheus • Grafana • DataDog • New Relic • SLO/SLI • Alerting • Troubleshooting

Security & Compliance: HashiCorp Vault • SOC2/SOX/GDPR/ISO27001/PCI-DSS • Network Security • Access Control • Secret Management

Disaster Recovery: RTO/RPO Planning • Failover Automation • Backup Validation • DR Testing • Multi-Region Redundancy

Linux Systems: Ubuntu/RHEL/Fedora/CoreOS/Debian • RPM/DEB Packaging • Koji • Network Configuration • Performance Tuning • Systems Administration

Languages & Scripting: Python • Bash • Go • Network Automation

Open Source & Presentations

- github.com/goozbach
- gitlab.com/users/goozbach/projects
- presentation.goozbach.com
- linkedin.com/in/goozbach

Further Experience

Find the remainder of my experience on [LinkedIn](#).